

NAYANTARA

Catharanthus roseus (formerly Vinca rosea)

Quick Facts

Scientific Name	Catharanthus roseus (formerly Vinca rosea)
Family	Apocynaceae (Dogbane family)
Common Names	Nayantara (Bengali, meaning 'star of the eye'), Periwinkle, Sadabahar (Hindi, meaning 'ever-blooming'), Barmasi (Hindi), Vinca, Madagascar Periwinkle, Old Maid, Cape Periwinkle
Native Origin	Madagascar; naturalised throughout tropical regions worldwide
Colors	Pink, deep rose, white, lavender, pale lilac, red, and bi-coloured (white with pink or red eye centre). Single and double-flowered forms exist in cultivated varieties.
Height	30 to 80 cm as an erect herb or sub-shrub. Some sprawling varieties trail along the ground.
Blooming Season	Year-round throughout India without interruption. The name Sadabahar (ever-blooming) and Barmasi (twelve months) in Hindi reflect this continuous flowering habit. The plant does not have a dormant or non-flowering period in tropical India.

Distribution in India

One of the most ubiquitous wild and garden plants in India. Found growing along roadsides, in wastelands, on railway embankments, in garden borders, and in pots virtually everywhere in the country from sea level to about 1,200 metres altitude. Particularly abundant in West Bengal (where it is called Nayantara and is culturally loved), Odisha, Andhra Pradesh, Tamil Nadu, Maharashtra, Kerala, and Uttar Pradesh. Flowers abundantly in the Kolkata urban landscape year-round.

Description and Overview

Catharanthus roseus, called Nayantara in Bengali and Sadabahar in Hindi, is a perennial flowering plant that is simultaneously one of India's most beloved garden flowers, most important medicinal plants, and most successful wild plants. The plant is technically native to Madagascar but has naturalised so thoroughly across tropical regions that it is now considered a cosmopolitan tropical species.

The flowers are five-petalled with a characteristic 'pinwheel' or 'propeller' arrangement. Each petal is broadly oval and connected at the base to form a narrow tube from which the petals flare outward. In the centre of the flower is a small eye, often of a contrasting colour. The name Nayantara (Bengali: eye-star, star of the eye) is believed to reference the eye-like centre of the flower.

From a medical and pharmaceutical standpoint, Catharanthus roseus is among the most globally important medicinal plants of the 20th century. In the 1950s, researchers investigating the plant's

traditional use for diabetes isolated two alkaloids — vincristine and vinblastine — that were found to have remarkable anti-cancer (cytotoxic) properties. Vincristine became a breakthrough chemotherapy drug for treating childhood leukaemia (acute lymphoblastic leukaemia), certain lymphomas, and other cancers. Vinblastine is used against Hodgkin's lymphoma. These drugs are included in the World Health Organisation's List of Essential Medicines and have saved millions of lives worldwide. The plant also contains vindesine and vinorelbine, other clinically important alkaloids.

In Indian folk medicine, the plant has a long history of use for diabetes, hypertension, and skin conditions, predating the scientific discovery of its alkaloids. In Ayurveda, the plant is used as a hypoglycaemic (blood sugar lowering) remedy.

In Bengali culture, the Noyantara flower holds a place of particular affection. It is used in informal garlands, offered at village shrines, and is a common motif in Bengali folk art, literature, and poetry. The name itself has been used as a given name for girls in Bengal.

Blooming Season and Conditions

Year-round throughout India without interruption. The name Sadabahar (ever-blooming) and Barmasi (twelve months) in Hindi reflect this continuous flowering habit. The plant does not have a dormant or non-flowering period in tropical India.

Cultivation

Catharanthus roseus is propagated by seeds or stem cuttings. Seeds germinate within 10 to 15 days at 22 to 28 degrees Celsius. In Indian gardens, the plant is grown as a perennial and rarely needs to be replanted — it self-seeds freely and reappears each year. It tolerates a wide range of soils including sandy, loamy, and even poor soils, but prefers well-drained conditions.

Commercial cultivars such as the 'Vinca' series and 'Catharanthus' hybrid series are widely sold in Indian nurseries in pink, white, red, and bicolor forms. These cultivars have been bred for larger flowers, more compact growth, and improved heat and humidity tolerance. The plant requires full sun for best flowering and is relatively drought-tolerant once established.

Uses and Applications

Pharmaceutical — cancer treatment: The primary global significance. Vincristine and vinblastine extracted from the plant are essential chemotherapy drugs used worldwide for leukaemia and lymphoma treatment.

Traditional medicine: Used across tropical Asia and Latin America for treating diabetes, hypertension, fever, sore throat, and skin disorders.

Ornamental: Among the most popular bedding and pot plants in Indian urban gardens due to its continuous blooming habit, heat tolerance, and minimal care requirements.

Companion planting: The plant's alkaloids have mild repellent effects on certain garden insects.

Interesting Facts

- The discovery of vincristine and vinblastine from this plant is one of the most significant achievements in ethnobotanical drug discovery of the 20th century.

- Before the development of vincristine, childhood acute lymphoblastic leukaemia had a survival rate of less than 10 percent. Today, with vincristine-based chemotherapy, the survival rate exceeds 90 percent.
- The global revenue from vincristine and vinblastine drugs runs into hundreds of millions of dollars annually, yet the country of origin (Madagascar) has historically received little benefit from this discovery — a case study in biopiracy debates.
- The plant can produce a new bloom cycle continuously throughout the year in tropical climates, making it one of very few plants that flower without any interruption.
- In West Bengal, the flower is so commonly associated with simplicity and accessible beauty that it appears in the poetry of Rabindranath Tagore and other Bengali literary figures.

Source: Flower AI Knowledge Base — NAYANTARA entry. Scientific name: *Catharanthus roseus* (formerly *Vinca rosea*). Family: Apocynaceae (Dogbane family).